

Tahoe Regional Planning Agency
Environmental Improvement Program
MEEKS MEADOW SEZ: CONIFER ENCROACHMENT ASSESSMENT

Introduction

Of the 10 Stream Environment Zone (SEZ) indicators identified by the TRPA Environmental Improvement Program (EIP) and defined in the SEZ Baseline Condition Assessment report (2020), the effects of *conifer encroachment* pose the greatest threat to the success of the Meeks Meadow Restoration Project. For meadow projects like the restoration of the four SEZ units of Meeks Bay Meadow, reversing *conifer encroachment* serves to eliminate excess wildfire fuels while maintaining the unique biodiversity and ecosystem functions of the historic meadow habitat.

Background

TRPA's SEZ Baseline Condition Assessment report (2020) provides a starting point and "baseline condition" from which future examinations into SEZ indicator health can be built. In Appendix 5, which focuses on conifer encroachment into meadows, the authors outline their use of historic aerial imagery, soil mapping, LiDAR data, and field verification to determine the extent of historic meadow SEZs. The section then details data processing and summarization methods for determining current levels of conifer encroachment with up-to-date LiDAR data.

Methods

This study replicated the LiDAR-based conifer encroachment assessment methodology set forth in the SEZ Baseline Condition Assessment report, with only slight deviations from the published procedure (TRPA 2020). The essential components of the TRPA methodology were maintained: data insights are summarized visually and arithmetically according to a vector-based, hexagonal grid of 300 square meter cells, which is overlaid atop the extent of the four SEZ units of Meeks Bay Meadow. Procedure steps are outlined narratively below, and in detailed pseudo-code in the **Technical Workflow**.

- 1) Map: Conifers 2009
Technical Workflow: *TRPA Data Acquisition*
- 2) Map: Conifers 2022
Technical Workflow: *LiDAR Data Acquisition, LiDAR Data Pre-processing*
- 3) Overlay 300 square meter hexagonal grid
Technical Workflow: *Summarizing with a Hexagonal Vector Grid I*
- 4) Calculate: Δ Encroachment, %Encroachment
Technical Workflow: *Summarizing with a Hexagonal Vector Grid II*

1. TRPA's vector dataset '*Trees_in_Meadow*', which was itself derived from LiDAR data from 2009, was utilized to identify conifer trees in Meeks Bay Meadow SEZ units for 2009. Any tree in the vector dataset with !MaxHt! attribute greater than or equal to 6 meters was considered a conifer.
2. Vector points for conifer trees in Meeks Bay Meadow SEZ units for 2022 had to be derived from current LiDAR data according to the procedure outlined by TRPA (2020, Appendix 5; USGS 2023). This study did not make use of Tree Approximate Objects analysis, rather opting to create a Digital Surface Model (DSM) and Digital Terrain Model (DTM) from LiDAR to calculate a canopy height raster. This canopy height raster was used in conjunction with focal statistics to identify tops of trees whose height was greater than 6 meters. These conifer treetops were identified in a binary raster which was then converted into vector points.
3. A vector-based hexagonal grid with cells of 300 square meters was overlaid atop the Meeks Bay Meadows SEZ units. A spatial join was used to count the number of conifer trees (2009 vector data, 2022 LiDAR-derived vector data) in each hexagon. A binary field for "Encroachment" was calculated where one or more conifers in a hexagon meant that the hexagon was encroached.
4. Δ Encroachment (Change) was calculated for each hexagon (see classification scheme in **Technical Workflow**). A spatial join was used to summarize percentages of SEZ units that fell into each Δ Encroachment category. Another spatial join was then used to summarize the percentage of each SEZ unit that was encroached, in order to determine ratings and scores for each unit (Table 1 & 2, Figure 1).

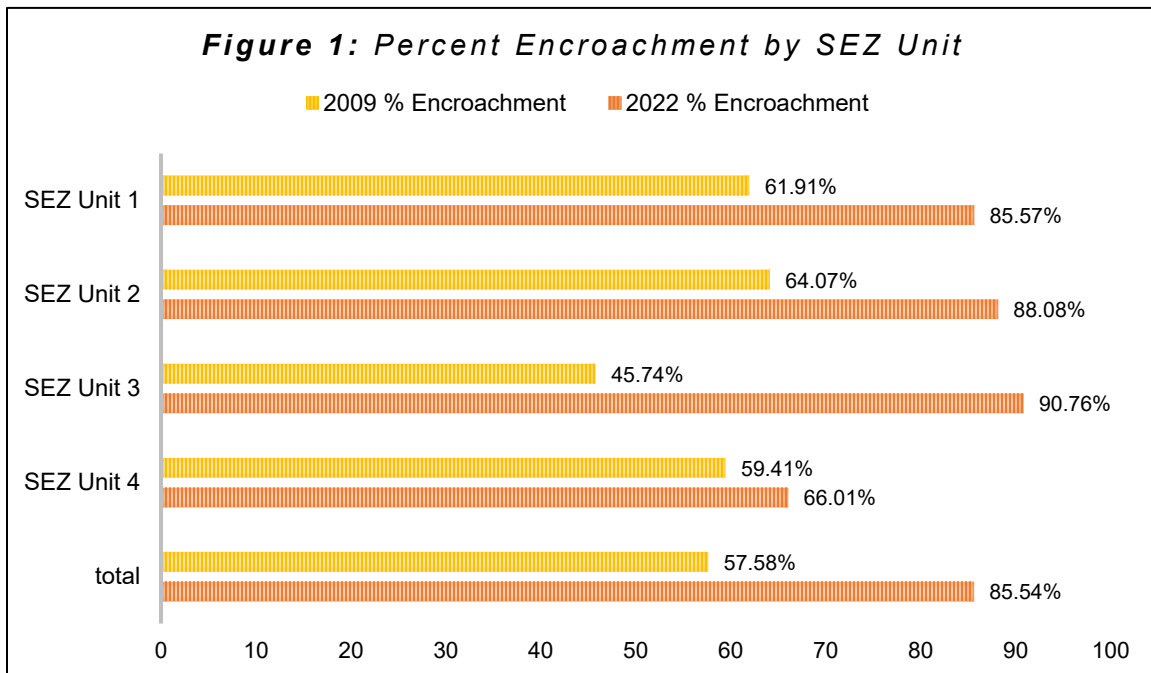
Tables and Figures

Table 1: Meeks Bay Meadow SEZ Units – Change in Encroachment: Cell Count (%/Total)

MBM SEZ Unit	No Encr. (%/Total)	Retreat (%/Total)	New Encr. (%/Total)	Ongoing Encr. (%/Total)	Total Cell Count
SEZ Unit 1	173 (13.05%)	11 (0.83%)	333 (25.11%)	809 (61.01%)	1326
SEZ Unit 2	61 (8.62%)	17 (2.40%)	190 (26.84%)	440 (62.14%)	708
SEZ Unit 3	72 (7.89%)	12 (1.31%)	394 (43.15%)	435 (47.65%)	913
SEZ Unit 4	102 (28.41%)	2 (0.56%)	26 (7.24%)	229 (63.79%)	359
total	408 (12.34%)	42 (1.27%)	943 (28.52%)	1913 (57.86%)	3306

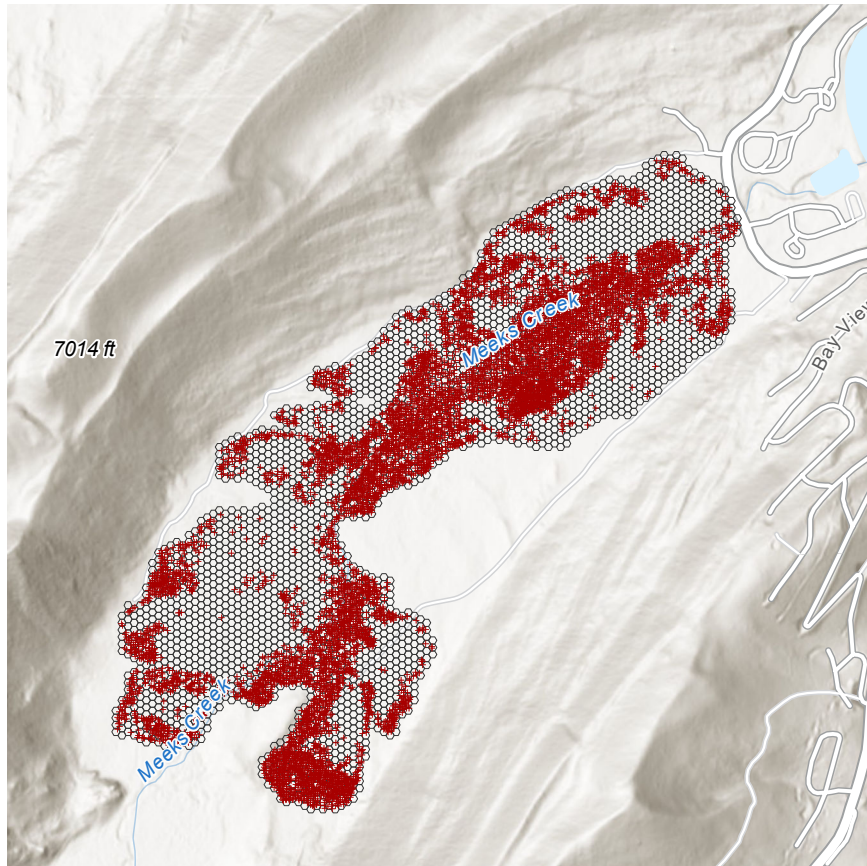
Table 2: Meeks Bay Meadow SEZ Units – Conifer Encroachment Indicator: Rating / Score

MBM SEZ Unit	2009 % Encroachment	2009 Indicator Rating / Score	2022 % Encroachment	2022 Indicator Rating / Score
SEZ Unit 1	61.91	C / 6	85.57	D / 3
SEZ Unit 2	64.07	C / 6	88.08	D / 3
SEZ Unit 3	45.74	C / 6	90.76	D / 3
SEZ Unit 4	59.41	C / 6	66.01	C / 6
total	57.58	C / 6	85.54	D / 3

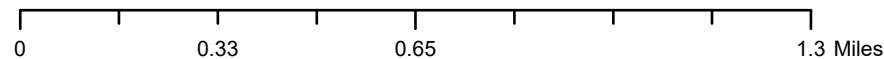
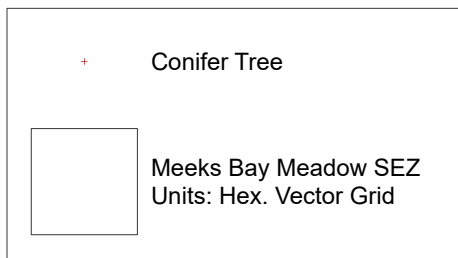
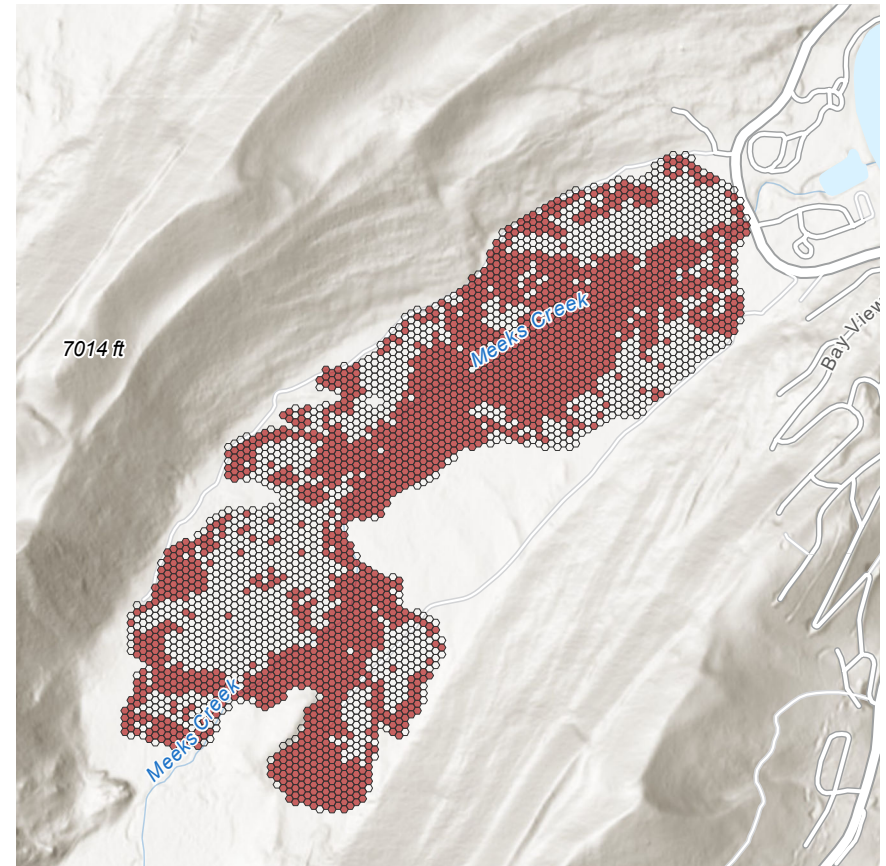


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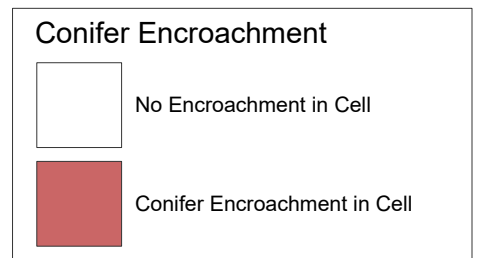
Map 1a. 'Trees in SEZ' (2009)



Map 1b. Conifer Encroachment (2009)



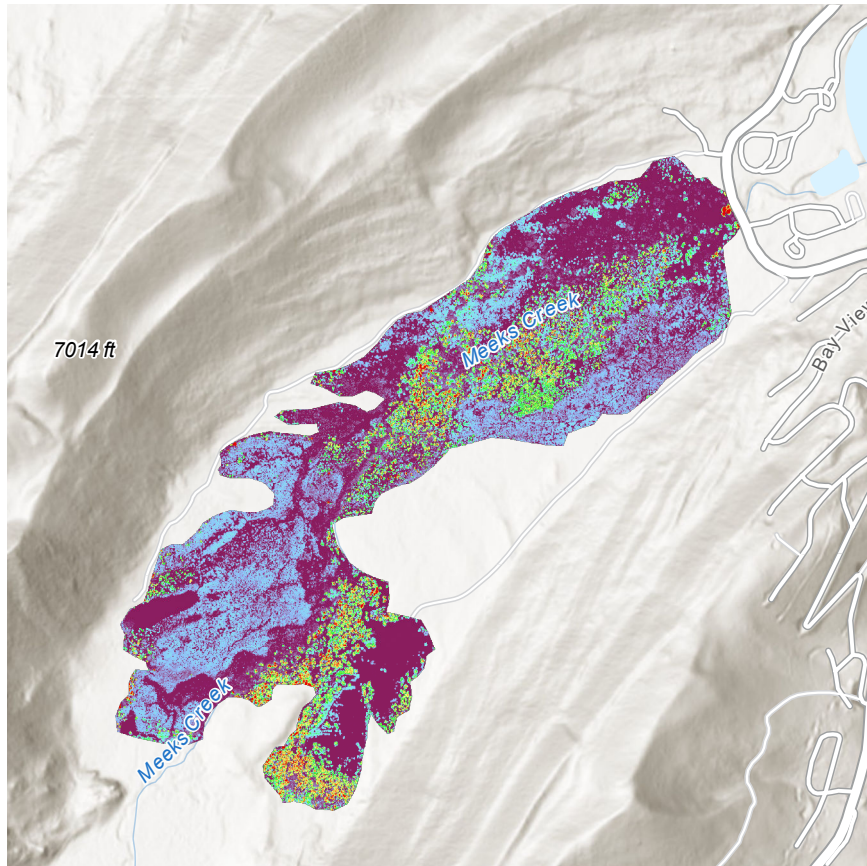
Scale: 1:20,000
PCS: NAD 1983 UTM Zone 10N
GCS: GCS North American 1983
Projection: Transverse Mercator



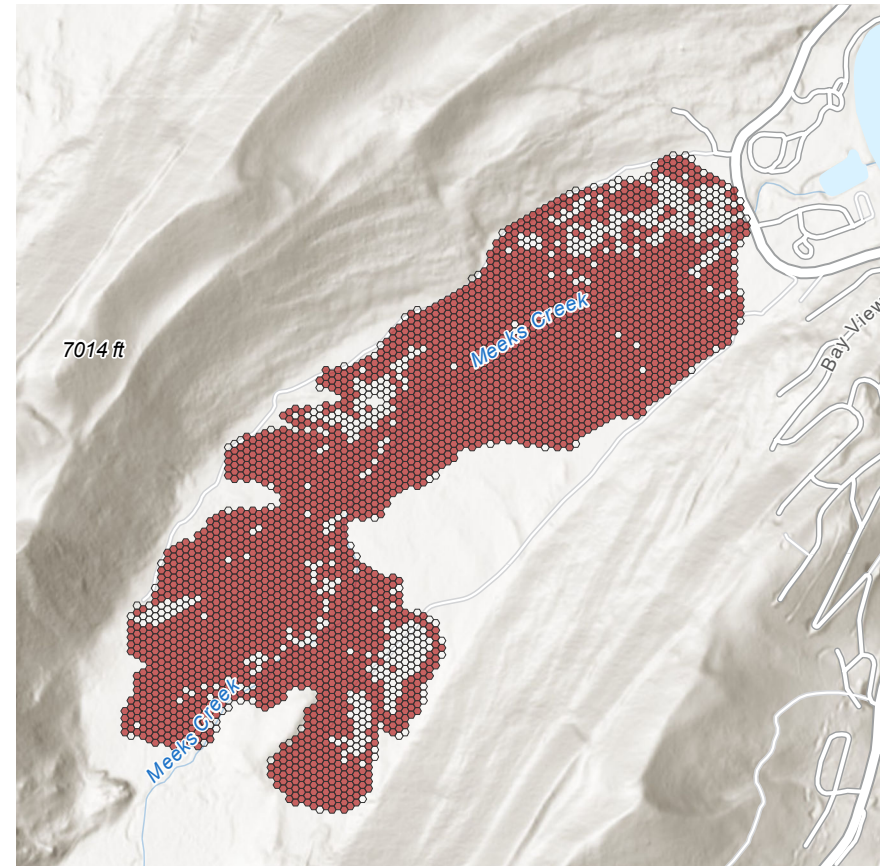
Conifer Tree data: Tahoe Regional Planning Agency. 2020. "Trees in Meadow" [Data set]. TRPA GIS. https://maps.trpa.org/server/rest/services/Trees_in_Meadow/MapServer
Basemaps: Esri, NASA, NGA, USGS, FEMA, Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community
Cartography, Analysis: Aaron Goodman, UCLA Geography
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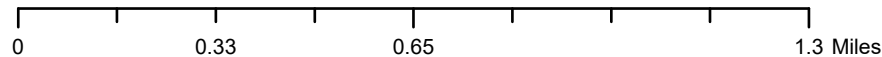
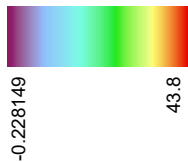
Map 2a. Canopy Height (2022)



Map 2b. Conifer Encroachment (2022)



2022 Canopy Height (meters)



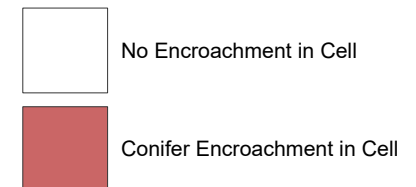
Scale: 1:20,000

PCS: NAD 1983 UTM Zone 10N

GCS: GCS North American 1983

Projection: Transverse Mercator

Conifer Encroachment



Canopy Height data from LiDAR: United States Geological Survey. 2023. "Sierra Nevada Work Unit 5, California: Lidar Technical Data" [Data set]. NV5, USGS. http://prd-tnm.s3.amazonaws.com/index.html?prefix=StagedProducts/Elevation/metadata/CA_SierraNevada_B22/CA_SierraNevada_5_2022

LiDAR Scenes: 10SGJ4723 <https://www.sciencebase.gov/catalog/item/66c987acd34e98e8a9240ec0>; 10SGJ4724 <https://www.sciencebase.gov/catalog/item/66c9a458d34e98e8a92428b0>
10SGJ4823 <https://www.sciencebase.gov/catalog/item/66c98575d34e98e8a9240ce2>; 10SGJ4824 <https://www.sciencebase.gov/catalog/item/66c993e4d34e98e8a9241988>

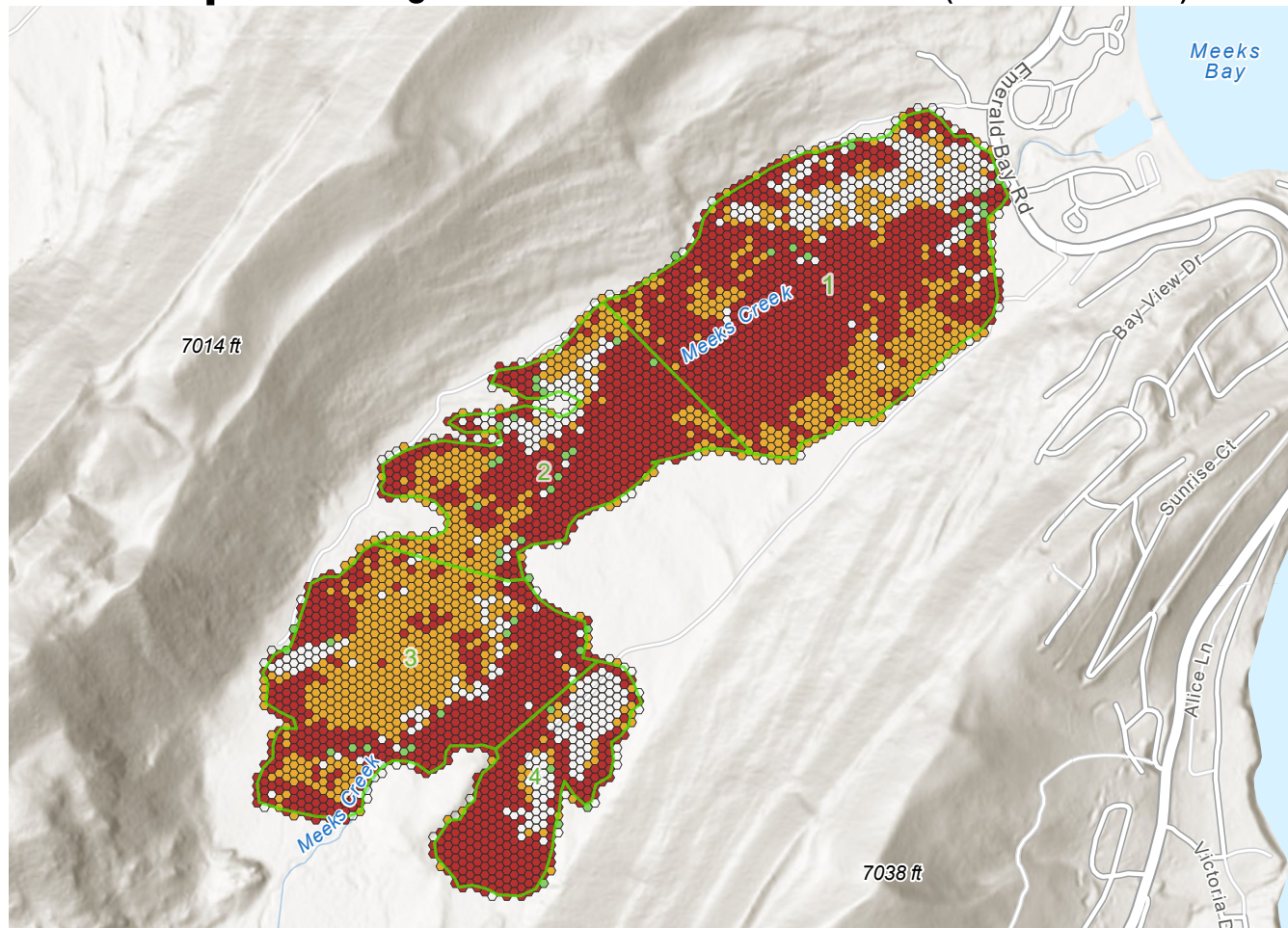
Basemaps: Esri, NASA, NGA, USGS, FEMA, Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

Cartography, Analysis: Aaron Goodman, UCLA Geography

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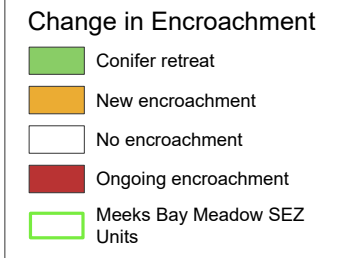
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Map 3. Change in Conifer Encroachment (2009 - 2022)



0 0.25 0.5 1 Miles

Scale: 1:16,000
PCS: NAD 1983 UTM Zone 10N
GCS: GCS North American 1983
Projection: Transverse Mercator



LiDAR Data: Tahoe Regional Planning Agency. 2024. "SEZ Assessment Unit" [Data set]. TRPA GIS. https://maps.trpa.org/server/rest/services/SEZ_Assessment_Unit/MapServer
Canopy Height data: United States Geological Survey. 2023. "Sierra Nevada Work Unit 5, California: Lidar Technical Data" [Data set]. NV5, USGS. http://prd-tnm.s3.amazonaws.com/index.html?prefix=StagedProducts/Elevation/metadata/CA_SierraNevada_B22/CA_SierraNevada_5_2022
LiDAR Scenes: 10SGJ4723 <https://www.sciencebase.gov/catalog/item/66c987acd34e98e8a9240ec0>; 10SGJ4724 <https://www.sciencebase.gov/catalog/item/66c9a458d34e98e8a92428b0>
10SGJ4823 <https://www.sciencebase.gov/catalog/item/66c98575d34e98e8a9240ce2>; 10SGJ4824 <https://www.sciencebase.gov/catalog/item/66c993e4d34e98e8a9241988>
Basemaps: Esri, NASA, NGA, USGS, FEMA, Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community
Cartography, Analysis: Aaron Goodman, UCLA Geography
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Discussion

This study revealed significant developments in conifer encroachment within Meeks Bay Meadow SEZ units between 2009 and 2022. In all four SEZ units, the majority of hexagonal cells exhibited Ongoing Encroachment over the study period (Table 1; Map 3). In SEZ Units 1, 2, and 3, the next most common Δ Encroachment development was New Encroachment, which is reflected in those SEZ units' demoted 2022 Indicator Ratings of "D". SEZ Unit 4, which did not exhibit as much New Encroachment as the other units, maintained its "C" rating for 2009 and 2022 Indicator analyses (Table 2). SEZ Unit 3 saw the most dramatic increase in conifer encroachment over the study period, from 45.74% to 90.76% of its cell area. SEZ Unit 3 had the lowest percentage of cells encroached in 2009, and the greatest percentage encroached in 2022.

These results have important implications for meadow restoration efforts. The spatial pattern of unchecked conifer expansion in SEZ Units 1, 2, and 3 (Ongoing Encroachment, New Encroachment) poses a priority for conifer removal projects. The relative stability in encroachment level exhibited in SEZ Unit 4 suggests environmental factors in that unit or ongoing management efforts have succeeded in resisting conifer encroachment there.

However, these data and methods come with some limitations that must be considered when interpreting results (TRPA 2020, Appendix 5, pp. 85).

In both 2009 and 2022 conifer vector datasets, the shared 6-meter height threshold poses limitations to our study's identification of conifer trees and encroachment. As pointed out by TRPA, young conifers below 6 meters in height are excluded from the analysis, which may lead to an underestimation of recent encroachment. This "blind spot" limits our ability to identify regions with recently established conifers that could benefit from early intervention. More rigorous object detection and deep learning models could improve future studies' ability to identify conifers with LiDAR, based on more parameters than canopy height.

Despite loose adherence to TRPA procedure in identifying conifer trees with LiDAR, this study's methods for processing trees into vector points for 2022 likely differed considerably from those used for the 2009 dataset (which were received in vector form). These methodological differences may manifest in systematic biases in tree detection. Considering this study's use of a simple canopy height raster with focal statistics, the results generated here for 2022 conifers are likely overestimates of tree count and conifer encroachment.

Finally, while the use of a hexagonal vector grid does provide a standardized method for data comparison, it does come with inherent spatial limitations. The 300 square meter cell size masks fine-scale patterns of conifer distribution within cells, and the

consolidation of conifer density into binary data further conceals variation in the encroachment data. In other words, for our study which fails to specify that some encroached cells only have one conifer, the intense encroachment of others is almost understated.

Overall, the LiDAR approach for conifer identification could be improved with more sophisticated object detection models that can account for tree shape and structure. Multispectral imagery analysis could offer another means by which the identification of conifers could be verified. Finally, the development of density-based metrics for encroachment in hexagonal cells could offer future studies greater insight into the intensity of encroachment across the meadow.

Despite the limitations of this study's methodologies, this assessment provides valuable insights into the trajectory of conifer encroachment in the Meeks Bay Meadow SEZ units. The prevalence of Ongoing and New conifer encroachment between 2009 and 2022 suggests that more aggressive management interventions may be needed to restore historic meadow habitat in Meeks Meadow.

Technical Workflow (1/4)

TRPA Data Acquisition

1. Add the following TRPA ArcGIS REST services to your ArcGIS Online account content, then import to ArcGIS Pro (ensuring editability):
 - `Trees_in_Meadow`
→ ``Trees_in_SEZ``
 - `SEZ_Assessment_Unit`
→ ``SEZ_Assessment_Unit``
2. Copy Features: input=``SEZ_Assessment_Unit`` w/ Definition query (`!Assessment_Unit_Name! LIKE '%Meeks Bay meadow%'`)
→ ``mbm_sez_units``
3. Copy Features: input=``Trees_in_SEZ`` w/ Selection by Location (Intersect ``mbm_sez_units``)
→ ``mbm_sez_trees_2009``
4. ``mbm_sez_trees_2009`` w/ Definition query (`MaxHt >= 6`)
→ ``mbm_sez_trees_2009``

LiDAR Data Acquisition

1. Download selected LiDAR scenes for 2022:
 - a. `wget`
https://rockyweb.usgs.gov/vdelivery/Datasets/Staged/Elevation/LPC/Projects/CA_SierraNevada_B22/CA_SierraNevada_5_2022/LAZ/USGS_LPC_CA_SierraNevada_B22_10SGJ4723.laz
 - b. `wget`
https://rockyweb.usgs.gov/vdelivery/Datasets/Staged/Elevation/LPC/Projects/CA_SierraNevada_B22/CA_SierraNevada_5_2022/LAZ/USGS_LPC_CA_SierraNevada_B22_10SGJ4724.laz
 - c. `wget`
https://rockyweb.usgs.gov/vdelivery/Datasets/Staged/Elevation/LPC/Projects/CA_SierraNevada_B22/CA_SierraNevada_5_2022/LAZ/USGS_LPC_CA_SierraNevada_B22_10SGJ4823.laz
 - d. `wget`
https://rockyweb.usgs.gov/vdelivery/Datasets/Staged/Elevation/LPC/Projects/CA_SierraNevada_B22/CA_SierraNevada_5_2022/LAZ/USGS_LPC_CA_SierraNevada_B22_10SGJ4824.laz
2. Use Python's `'laspy'` package to read the `.laz` files and rewrite them as uncompressed `.las` files
 - a.

```
```python
import laspy
Read LAZ file
las_data = laspy.read("input.laz")
Write to LAS format
las_data.write("output.las")
```
```
3. Create an `.lasd` file in ArcGIS Pro with connections to your four newly created `.las` files
→ ``mbm_sez_lidar_LASD.lasd``

Technical Workflow (2/4)

LiDAR Data Pre-processing

DSM Calculation

1. Use ArcGIS Pro's "LAS Filters" setting to toggle to "First Return".
2. LAS Dataset to Raster: input=`mbm\_sez\_lidar\_LASD.lasd`, value=Elevation, interpolation type=Binning, cell assignment=Maximum, void fill=Linear, output data type=Floating point, sampling type=Cell Size, sampling value=0.5
 ### Consideration for sampling value: 151,374,554 pts/4,000,000 m²
 ### = ~38 points per cell; very dense
 → `mbm\_sez\_dsm`

DTM Calculation

1. Use ArcGIS Pro's "LAS Filters" setting to toggle to "Ground".
2. LAS Dataset to Raster: input=`mbm\_sez\_lidar\_LASD.lasd`, value=Elevation, interpolation type=Binning, cell assignment=Minimum, void fill=Linear, output data type=Floating point, sampling type=Cell Size, sampling value=0.5
 ### Consideration for sampling value: 151,374,554 pts/4,000,000 m²
 ### = ~38 points per cell; very dense
 → `mbm\_sez\_dtm`

Canopy Height Calculation

1. Raster Calculator: `mbm\_sez\_dsm` - `mbm\_sez\_dtm` =
 → `mbm\_sez\_canopyHeight`

Clip Raster

1. Clip Raster: input=`mbm\_sez\_canopyHeight`, output extent=`mbm\_sez\_units`, Use Input Features for Clipping Geometry
 → `mbm\_sez\_canopyHeight\_Clip`

Focal Statistics

1. Focal Statistics: input=`mbm\_sez\_canopyHeight\_Clip`, neighborhood=Circle, width=height=5, units type=Cell, statistics type=Maximum
 → `mbm\_sez\_canopyHeight\_Clip\_focalStat`
 ### Using this Focal Statistics operation helps reduce the number of
 ### cells to be passed through the next Raster To Point operation

Raster To Point

1. Raster Calculator: input=`mbm\_sez\_canopyHeight\_Clip\_focalStat`,
 con((`mbm\_sez\_canopyHeight\_Clip`==`mbm\_sez\_canopyHeight\_Clip\_focalStat`)
) & (`mbm\_sez\_canopyHeight\_Clip`>=6),1,0)
 → `mbm\_sez\_conifers\_Clip\_focalStat`
 ### This Raster Calculator expression identifies local maxima or tops
 ### of trees with the use of the _focalStat layer created above, while
 ### still restricting the search to trees above 6m
2. Raster To Point: input=`mbm\_sez\_conifers\_Clip\_focalStat`
 → `mbm\_sez\_trees\_2022\_focalStat`

Technical Workflow (3/4)

Summarizing with a Hexagonal Vector Grid I

Grid Set-up

1. Generate Grids and Hexagons:
area of interest=`mbm\_sez\_units`, cell geometry type=Hexagon,
cell size=300 Square Meters
➔ `mbm\_sez\_hexGrid`
This Raster Calculator expression identifies local maxima or tops
of trees with the use of the `_focalStat` layer created above, while
still restricting the search to trees above 6m
2. Create Field(s): `mbm\_sez\_hexGrid`
!count2009!, !count2022!, !bin2009!, !count2022!, !changeEncr!

2009 Data Import

1. Spatial Join:
target features=`mbm\_sez\_hexGrid`
join features=`mbm\_sez\_trees\_2009`
Keep All Target Features
match option=Contains
2. Calculate Field: `mbm\_sez\_hexGrid` : ➔!count2009!
!count2009! == !Join_Count!
3. Calculate Field: `mbm\_sez\_hexGrid` : ➔!bin2009!
!bin2009! == 1 if !count2009! >= 1 else 0
4. (remove Spatial Join)

2022 Data Import

1. Spatial Join:
target features=`mbm\_sez\_hexGrid`
join features=`mbm\_sez\_trees\_2022\_focalStat`
Keep All Target Features
match option=Contains
2. Calculate Field: `mbm\_sez\_hexGrid` : ➔!count2022!
!count2022! == !Join_Count!
3. Calculate Field: `mbm\_sez\_hexGrid` : ➔!bin2022!
!bin2022! == 1 if !count2022! >= 1 else 0
4. (remove Spatial Join)

Calculating Change in Encroachment

1. Calculate Field: `mbm\_sez\_hexGrid` : ➔!changeEncr! (Text)
```{python}  
def getChange(bin2009, bin2022):  
 if bin2009==0 and bin2022==0 # no conifers in either period  
 return 'No encroachment'  
 if bin2009==1 and bin2022==0 # conifers present 2009, absent 2022  
 return 'Conifer retreat'  
 if bin2009==0 and bin2022==1 # conifers absent 2009, present 2022  
 return 'New encroachment'  
 if bin2009==1 and bin2022==1 # conifers present in both periods  
 return 'Ongoing encroachment'  
getChange(!bin2009!, !bin2022!)  
```

Technical Workflow (4/4)

Summarizing with a Hexagonal Vector Grid II

Generating Table 1: Change in Encroachment (Cell Count, %/Total)

1. Spatial Join:
 target features=`mbm\_sez\_hexGrid`
 join features=`mbm\_sez\_units`
 Keep All Target Features
 match option=Have their center in
2. Summary Statistics: input table=`mbm\_sez\_hexGrid` x `mbm\_sez\_units`,
 statistics fields=!OBJECTID!, statistics type=COUNT, (filler)
 case fields=!changeEncr!, !Assessment Unit Name!
 → `mbm_sez_hexGrid_x_units_Stats` (TABLE)
 ### Table `mbm_sez_hexGrid_x_units_Stats` lists count of hexagonal cells
 ### for each combination of !changeEncr! category with SEZ Unit via the
 ### !Assessment Unit Name! attribute
3. (copy results (counts) into external spreadsheet editor, or continue with calculations in ArcGIS Pro)
 a. $\% / \text{Total} = (\text{Unit Cell Count (in category)} / \text{Total Cell Count}) * 100$
4. (remove Spatial Join)

Generating Table 2: Conifer Encroachment Indicator (Rating/Score)

1. Spatial Join:
 target features=`mbm\_sez\_units`
 join features=`mbm\_sez\_hexGrid`
 Keep All Target Features
 match option=Contains
 field map={COUNT(!GridID!), SUM(!bin2009!), SUM(!bin2022!)}
 ### `mbm\_sez\_units` now has total counts of hexagonal cells for each
 ### unit in the !GridID! field, and counts of cells with 1's in the
 ### !bin2009! and !bin2022! fields
2. (copy results (counts) into external spreadsheet editor, or continue with calculations in ArcGIS Pro)
 a. $2009\% \text{Encroachment} = (2009 \text{ Encr. Cell Count} / \text{Total Cell Count}) * 100$
 b. $2022\% \text{Encroachment} = (2022 \text{ Encr. Cell Count} / \text{Total Cell Count}) * 100$
 c. 2009/2022 Rating/Score: see classification scheme in **Methods** or TRPA
 2020 Baseline Condition Assessment
3. (remove Spatial Join)

Works Cited

Literature

1. Tahoe Regional Planning Agency. 2020. *Baseline Condition Assessment*. TRPA GIS, EPA.
gis.trpa.org/TahoeSEZViewer/SEZ%20baseline%20condition%20assessment_v8.pdf

Datasets

1. Tahoe Regional Planning Agency. 2020. *Trees_in_Meadow* [Data set]. TRPA GIS. maps.trpa.org/server/rest/services/Trees_in_Meadow/MapServer
2. Tahoe Regional Planning Agency. 2024. *SEZ_Assessment_Unit* [Data set] TRPA GIS. maps.trpa.org/server/rest/services/SEZ_Assessment_Unit/MapServer
3. United States Geological Survey. 2023. *Sierra Nevada Work Unit 5, California: Lidar Technical Data* [Data set]. NV5, USGS. prd-tnm.s3.amazonaws.com/index.html?prefix=StagedProducts/Elevation/metadata/CA_SierraNevada_B22/CA_SierraNevada_5_2022

Specific LiDAR Scenes

1. 10SGJ4723
<https://www.sciencebase.gov/catalog/item/66c987acd34e98e8a9240ec0>
2. 10SGJ4724
<https://www.sciencebase.gov/catalog/item/66c9a458d34e98e8a92428b0>
3. 10SGJ4823
<https://www.sciencebase.gov/catalog/item/66c98575d34e98e8a9240ce2>
4. 10SGJ4824
<https://www.sciencebase.gov/catalog/item/66c993e4d34e98e8a9241988>